

ABISOLA Z. KEHINDE

Cancer Biology | Metabolic Dependencies | Multi-omics | Precision Oncology

Vancouver, BC, Canada | abisola.kehinde@ubc.ca | +1 250 808 3712 | linkedin.com/in/abisolazkehinde | ORCID 0000-0002-9916-3058

RESEARCH PROFILE

PhD Candidate in Biochemistry and Molecular Biology and Vanier Canada Graduate Scholar investigating targetable metabolic dependencies in pancreatic cancer. My research combines inducible perturbation systems, mammalian cell models, LC-MS metabolomics, stable-isotope tracing, enzyme assays, molecular biology, and multi-omics to define mechanisms that limit cancer-cell adaptation. I am particularly interested in how tumour-specific dependencies can be translated into biomarkers and therapeutic strategies, and in extending this framework to DNA-damage response, synthetic lethality, and treatment resistance in breast cancer.

RESEARCH EXPERTISE

Cancer metabolism and tumour dependencies • Mechanistic cell biology • Target validation concepts • Mammalian cell culture and pancreatic cancer organoids • Inducible lentiviral perturbation systems • Cloning, qPCR and western blotting • Flow cytometry • Proliferation and clonogenic assays • Enzyme assays • Drug/nutrient perturbation and rescue studies • LC-MS metabolomics • Stable-isotope tracing • RNA-seq, ATAC-seq and ribosome profiling • R, Python and shell • Experimental design and quantitative data interpretation

RESEARCH EXPERIENCE

Graduate Researcher | Parker Lab, University of British Columbia / BC Children's Hospital Research Institute | Vancouver, Canada | 2021–Present

- Lead a mechanistic cancer-metabolism project asking how acute alanine restriction creates targetable metabolic liabilities in pancreatic cancer cells, integrating perturbation biology with metabolic and growth phenotyping.
- Engineered and validated a doxycycline-inducible alanine-depletion system and used nutrient rescue, inhibitor studies, purified-enzyme assays, and longitudinal cell-growth experiments to separate adaptive pathway changes from growth-limiting effects.
- Identified a serine-linked metabolic rerouting phenotype associated with GPT2 activity and hydroxypyruvate/glycerate accumulation, generating a mechanistic model of how altered substrate availability exposes a cancer-cell vulnerability.
- Use mammalian cancer-cell culture, pancreatic cancer organoids, lentiviral systems, molecular cloning, qPCR, western blotting, flow cytometry, proliferation/clonogenic assays, and biochemical assays to test pathway-level hypotheses.
- Develop targeted and untargeted LC-MS and stable-isotope-tracing workflows on an Orbitrap Exploris 240; perform sample preparation, calibration, system-suitability checks, peak review, troubleshooting, and quantitative interpretation.
- Integrate metabolomics with RNA-seq, ATAC-seq, and ribosome-profiling datasets to connect metabolic phenotypes with transcriptional, chromatin, and translational responses.
- Built reproducible R, Python, and shell workflows for normalization, statistical analysis, visualization, and multi-omics integration; developed a time-aware normalization framework for longitudinal metabolomics experiments.
- Present work at national and international meetings, prepare first-author manuscripts, and mentor junior researchers in experimental design, data interpretation, and scientific communication.

Graduate Researcher | PlantSMART Lab, University of British Columbia | Kelowna, Canada | 2019–2021

- Investigated biochemical and compositional phenotypes using microscopy, spectroscopy, enzymatic assays, and quantitative analysis; contributed to experimental design, manuscript preparation, and peer-review responses.
- Led the writing and revision of a first-author study on dietary starch, protein, and fibre composition across breadfruit cultivars and hybrids.

EDUCATION

PhD, Biochemistry and Molecular Biology | University of British Columbia, Vancouver, Canada | Expected 2026

MSc, Biochemistry and Molecular Biology | University of British Columbia, Kelowna, Canada | 2021 | Coursework average: 91.8%

BSc, Biochemistry, First Class Honours | University of Lagos, Lagos, Nigeria | 2016 | GPA 4.91/5.00

PhD thesis focus: metabolic vulnerabilities and adaptive nutrient metabolism in pancreatic cancer. Supervisor: Dr Seth J. Parker.

MSc thesis: Development of Novel Bioproducts from Breadfruit.

SELECTED PUBLICATIONS AND MANUSCRIPTS

- Kehinde AZ, Parker SJ, et al. Alanine restriction unmasks a GPT2-driven serine-to-hydroxypyruvate pathway that limits cell proliferation. First-author manuscript in preparation.
- Kehinde AZ, Parker SJ, et al. Time-aware normalization framework for longitudinal LC-MS metabolomics. First-author manuscript in preparation.
- Johal AS, Al-Shekaili HH, Abedrabbo M, Kehinde AZ, et al. Restricting lysine normalizes toxic catabolites associated with ALDH7A1 deficiency in cells and mice. *Cell Reports*. 2024;43(12):115069.
- Kehinde AZ, Parker SJ. No Back-up Plan: Loss of Isozyme Diversity as a Promising Therapeutic Strategy for Cancer. *Cancer Research*. 2022. First author.
- Kehinde AZ, Erland L, et al. South Pacific cultivars of breadfruit and their hybrids have unique dietary starch, protein and fiber. *Journal of Food Composition and Analysis*. 2022;105:104228. First author.
- Erland L, Needham A, Kehinde AZ, et al. Impact of microclimate on breadfruit fruit and nutritional quality. *Journal of Food Composition and Analysis*. 2023;115:104983.

SELECTED AWARDS, PRESENTATIONS AND LEADERSHIP

- Vanier Canada Graduate Scholarship (\$150,000 CAD), 2023–2026; UBC Four Year Doctoral Fellowship; UBC President's Academic Excellence Initiative PhD Award.
- Invited Speaker, Gordon Research Seminar (2025) and Canadian Society for Molecular Biosciences (2024); Best Talk Prize, UBC BMB Graduate Student Symposium (2024).
- Chair and Scientific Training Lead, BC Children's Hospital Research Institute Trainee Omics Group, 2023–2026; led workshops and seminars across metabolomics, genomics, bioinformatics, and high-performance computing.
- Co-led a successful \$25,000 CAD grant supporting Indigenous youth mentorship and equity-focused scientific training.